Problem 00000002692: a contraction with empty basin boundary

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Result and scope. The stated boundary identity is false for the polynomial

$$f(x) = 5x.$$

We use the usual convergence basin of the attracting fixed point 0,

$$\mathcal{A}_X(0) = \{x \in X : f^n(x) \longrightarrow 0\}, \qquad X = \mathbb{Q}_5 \text{ or } \mathbb{Z}_5.$$

This definition includes the fixed point itself, whose constant orbit converges to it. It directly implements the source's definition through iterative convergence on p-adic balls. We prove that the basin is the whole phase space, the periodic-point set is exactly $\{0\}$, and the basin boundary is empty. The unit-ball version also has empty boundary in the ambient field, so either the relative or the ambient topology gives the same counterexample.

Only the boundary clause is needed: the conjecture asserts a measure formula *and* this boundary identity, so a counterexample to the latter refutes the conjunction. We do not assign a multiplicative order to a nonunit residue, or claim a disproof of a separately repaired measure-only statement. There is no printed degree-at-least-two or invertible-multiplier hypothesis. The linear polynomial used here is nonconstant, has nonzero derivative, and is a genuine strict contraction.

1. A genuine 5-adic attracting system

Normalize the absolute value by $|5|_5 = 1/5$. The polynomial belongs to both $\mathbb{Q}_5[T]$ and $\mathbb{Z}_5[T]$. Its derivative is $f'(x) = 5 \neq 0$, in particular $|f'(0)|_5 < 1$. For all $x, y \in \mathbb{Q}_5$,

$$|f(x) - f(y)|_5 = \tfrac{1}{5}|x - y|_5.$$

The same identity holds on $\mathbb{Z}_5$, which is preserved by f. Thus both actual maps are contractions, with an attracting fixed point at 0.

Induction gives $f^n(x) = 5^n x$. Since $(1/5)^n |x|_5 \to 0$, every orbit converges to 0. Consequently

$$\mathcal{A}_{\mathbb{Q}_5}(0) = \mathbb{Q}_5, \qquad \mathcal{A}_{\mathbb{Z}_5}(0) = \mathbb{Z}_5.$$

Lean proves the actual iterate formula and convergence for multiplication by any element of norm less than one in a normed ring, and then specializes to Mathlib's actual types `Padic 5` and `PadicInt 5`. No finite residue-ring dynamics stands in for this convergence statement.

2. All periodic points and the actual boundary

Suppose that x is periodic with positive period n. Then $5^n x = x$, so

$$|x|_5 = |5^n x|_5 = (1/5)^n |x|_5.$$

Because $(1/5)^n < 1$, this forces $|x|_5 = 0$, hence $x = 0$. Conversely 0 is fixed. Thus the complete set of periodic points, allowing every positive period, is

$$\mathrm{Per}(f) = \{0\}.$$

The singleton is closed in either metric space. Since the whole space has empty frontier,

$$\partial_X \mathcal{A}_X(0) = \varnothing \quad \neq \quad \overline{\mathrm{Per}(f)}^{\,X} = \{0\}.$$

This uses the topological frontier, not a selected norm shell. Lean uses the existing periodic-set definition,

`Function.periodicPts`,

and the existing topological operations `frontier` and `closure`.

3. The unit ball inside the field

Let $\iota : \mathbb{Z}_5 \hookrightarrow \mathbb{Q}_5$ be the actual inclusion and set

$$B = \{x \in \mathbb{Q}_5 : |x|_5 \leq 1\}.$$

The inclusion commutes with the two maps, and Lean proves

$$\iota(\mathcal{A}_{\mathbb{Z}_5}(0)) = B.$$

The ball B is clopen for the non-Archimedean metric. Hence its ambient frontier is empty as well, whereas

$$\overline{\iota(\mathrm{Per}(f|_{\mathbb{Z}_5}))}^{\mathbb{Q}_5} = \{0\}.$$

This closes a possible distinction between computing the boundary in the ball and computing it in the ambient field. The standard identification of $\mathbb{Z}_p$ as the compact open unit ball of $\mathbb{Q}_p$ is also described in [1], p. 2; the Lean proof uses the actual subtype and its norm property.

For completeness, for every measure μ on the integer phase space, set equality gives

$$\mu(\mathcal{A}_{\mathbb{Z}_5}(0)) = \mu(\mathbb{Z}_5).$$

No normalization, numerical basin-volume formula, or order of 5 mod 5 is assumed. If normalized Haar measure is chosen, this simply says that the basin has the full unit-ball mass. The argument does not replace the basin by its punctured version $\mathcal{A}_X(0) \setminus \{0\}$; that is a different set and is not the convergence basin defined above.

4. Correction and attribution of the earlier submission

Closed PR 276 proposed $5x + x^2$ on $\mathbb{Z}_5$. The maintainer's review rejected its Lean proof because it formalized only elementary integer arithmetic and finitely many evaluations modulo 25, with none of the actual basin, measure, boundary, or periodic-point objects. Such computations do not establish 5-adic convergence or quantify all positive periods.

There was also a mathematical boundary error in its prose: the claimed basin $5\mathbb{Z}_5$ is clopen, so the shell $5\mathbb{Z}_5 \setminus 25\mathbb{Z}_5$ is not its topological boundary. Mapping that shell to 0 *modulo* 25 does not mean mapping it to 0 in $\mathbb{Z}_5$; for example $(5x+x^2)|_{x=5} = 50 \neq 0$. An undefined multiplicative order of 0 mod 5 is not assigned a numerical denominator here. The present proof uses a simpler nonzero linear multiplier and replaces every required dynamical and topological step with a theorem about the actual 5-adic objects. It does not rely on, or resubmit, the earlier finite computation.

5. Formal verification

The pinned Lean project includes the complete proof, an audit of every named declaration, and source-bound build and full fresh-kernel receipts. There is no `sorry`, `native_decide`, or additional axiom. The only admitted foundational dependencies are the standard `propext`, `Classical.choice`, and `Quot.sound`, when used. The actual polynomial evaluation, nonzero derivative, strict contraction, convergence basin, full periodic set, relative boundary, and ambient unit-ball boundary are formalized.

The complete original bilingual source is preserved in `SOURCE.md`. Worker and supervising AI review the mathematical scope and rendered pages; organizer acceptance is not asserted. The external reference is used only for the standard unit-ball convention, not for its later degree-at-least-two decomposition theorem. This result is selective production and supplies no new controlled efficiency measurement.

References

[1] S. Fan and L. Liao, *Dynamics of the square mapping on the ring of p-adic integers*, arXiv:1408.4574, 2014, p. 2. arXiv record.